Central Computers (Mod 6/2049) – General Data

1. General

The Central Computer is built on the Honeywell Primus APEX platform. The Central Computer systems architecture is built on modular hardware located in the aircraft and operational software. The Central Computer systems hardware is comprised of the following components:

- Modular Avionics Unit, MAU
- Data Acquisition Units, DAU 1 and DAU 2
- KDU-1080 or KDU-1080-2 Display Units – DU1, DU2, DU3, DU4
- KMC 2210 PFD Controllers
- KMC 2220 MFD Controller
- Display Reversionary Panel
- KMA 29 Audio Panels
- KMC 9200 Flight Controller

Data is communicated and received between the system along Avionic Standard Communication Bus, version D, (ASCB-D) are used to transfer data types between the MAU and the DAU and other peripherals. Maintenance data and software updates are transferred via the local area network, LAN.

2. Description

A. Modular Avionics Unit

The Modular Avionics Unit, MAU, with the modules installed acts like the central processing unit for Central Computer System. The MAU and modules, collect, compile and formats the data for the required system display. (Refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

B. Data Acquisition Unit

The Data Acquisition Units, DAU 1 and DAU 2 receives the analogue input signals from the aircraft sensors and transmitters, converts the analogue signals to a digital and send the digital signals to the Modular Avionics Unit, MAU to the Flight Data Recorder. (Refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data).

C. Display Units

The data is displayed on one of the four display units, DU1, DU2, DU3 and DU4, installed in the cockpit; two designated as the Primary Flight Displays, PFD, DU1, DU4 and two Multi-Function Displays, MFD, DU2, DU3. Under normal operations each Display Unit has a default format. (Refer to 31-11-00, Display Units – General Data).

D. Primary Flight Display Controllers

There are two KMC 2210 PFD Controllers, one for the Pilot's PFD one for the Co-pilot's PFD and is installed on the instrument panel to the right of the Pilot's PFD and to the left of the Co-pilot's PFD respectively. It allows the operator to navigate within the windows on the PFD screen and set the format for the system information to be displayed. (Refer to 31-11-00, Display Units – General Data).

E. Multi-Function Display Controller

The KMC 2220 multi-function controller is mounted on the aft face of the central control yoke in the cockpit. The multi-function controller provides the operator key access to menus, functions and formats for the information within each defined windows displayed on the MFDs. (Refer to 31-11-00, Display Units – General Data).

F. Display Reversionary Panel

The display system allows the pilot to revert the Display Units (DU) and Advanced Graphics Module 1 and 2 (AGM) in the event of failure of an individual display unit or AGM. The Display Reversion control panel is installed directly above the MFD controller, on the aft face of the control column.

G. Audio Panel

The KMA 29 audio control panels provide the pilot and co-pilot with monitoring of all the communication and navigation receivers, and microphones transmission over the selected radio transmitter. (Refer to 23-50-00, Audio Integration – General Data).

H. Flight Controller

The KMC 9200 Flight Controller is used to enter selected minima, set heading or track, and enter selected altitude.

I. ASCB–D Data Bus

Data is communicated and received between the system along Avionic Standard Communication Bus, version D, (ASCB–D) are used to transfer data types between the MAU and the DAU and other peripherals.

J. Local Area Network, LAN

The LAN is a standard general purpose data bus which is used for transferring maintenance data between components located on the LAN. The LAN also is used to loaded software updates via the aircraft maintenance panel located below the MAU in the Forward Avionics Compartment. (Refer to 45-00-00, Central Maintenance Computer – General data).